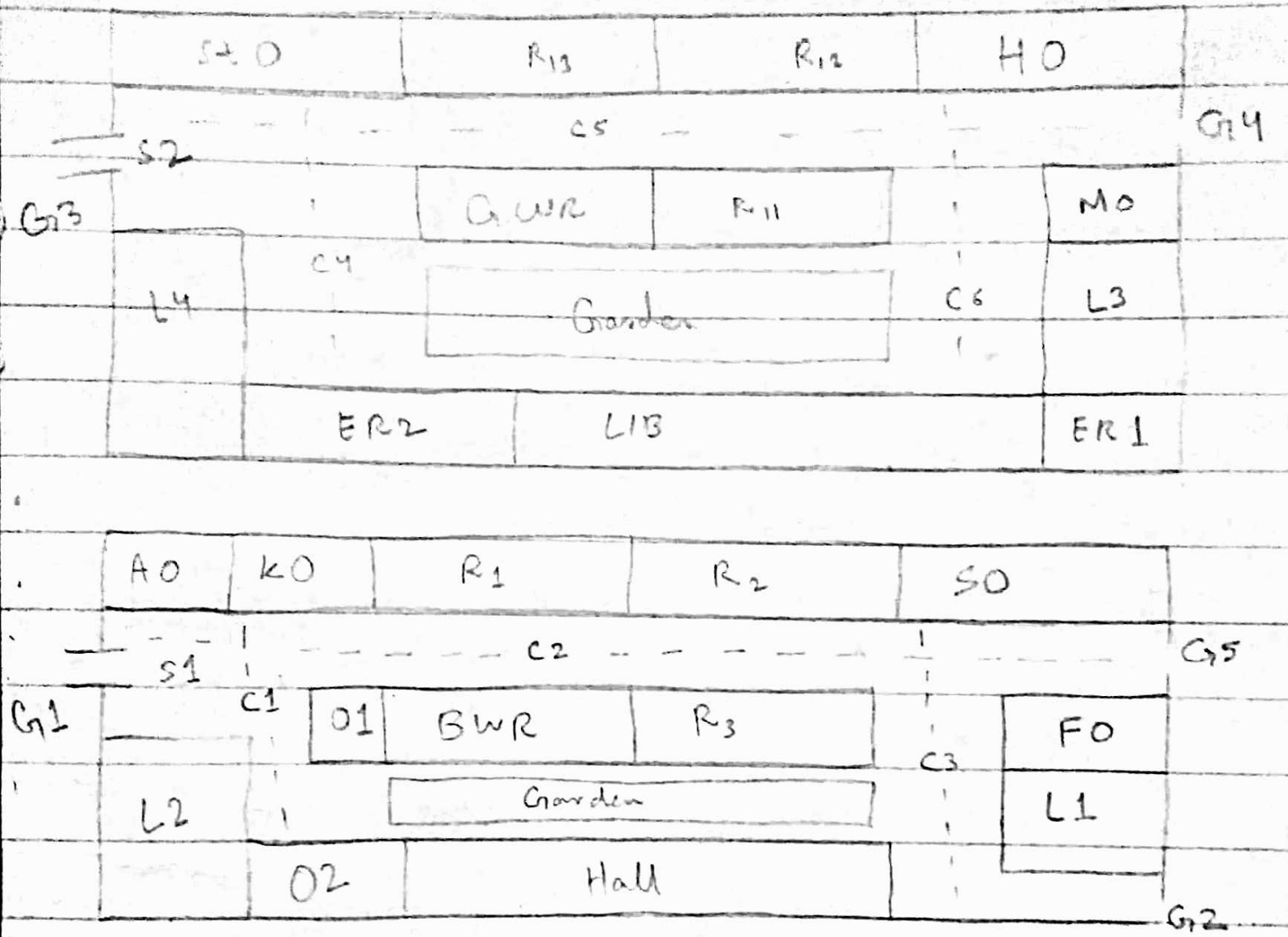
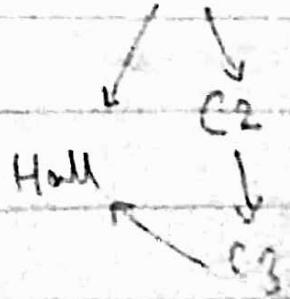
AI ⟷ Mahnoor BSCS 6B



Library to Hall Path

LIB → ER1 → C6 → C5 → C4 → S2 → S1 → C1



Initial State → Library Path cost → d : 1

Goal State → Hall State space → CS depart

Actions → movement in depart to reach hall

Transition Model →

LIB → ER 1

ER 1 → C 6

C 6 → C 5

C 5 → C 4

C 4 → S 2

S 2 → S 1

S 1 → C 1

C 1 → Hall , C 2

C 2 → C 3

C 3 → Hall

Shortest Path using BFS

Step 1 : Initialize

Q → [LIB]

V → [LIB]

Step 2 : dequeue LIB and Expand

Q → [ER1]

V → [LIB , ER1]

Step 3 : dequeue ER1 and Expand

Q → [C_6]

V → [LIB , ER1 , C_6]

Step 4 : dequeue C_6 and expand

Q → [C_5]

V → [LIB , ER1 , C_6 , C_5]

Step 5 : dequeue C_5 and expand

Q → [C_4]

V → [LIB , ER1 , C_6 , C_5 , C_4]

Step 6 : dequeue C_4 and Expand

Q → [S_2]

V → [LIB , ER1 , C_6 , C_5 , C_4 , S_2]

Step 7 : dequeue S_2 and Expand

Q → [S_1]

V → [LIB , ER_1 , C_6 , C_5 , C_4 , S_2 , S_1]

Step 8 : dequeue S_1 and Expand

Q → [C_1]

V → [LIB , ER_1 , C_6 , C_5 , C_4 , S_2 , S_1 , C_1]

Step 9 : dequeue C_1 and Expand

Q → [Hall , C_2]

V → [LIB , ER_1 , C_6 , C_5 , C_4 , S_2 , S_1 , C_1 , Hall]

Shortest Path is

LIB → ER_1 → C_6 → C_5 → C_4 → S_2 → S_1 → C_1 → Hall